REB, The Course
Class 12 Practice Assignment Solution

Liquid phase reaction (1) takes place in an adiabatic CSTR with a volume of 1.5 L operating at atmospheric pressure. The 50 °C feed to the reactor is 1.5 M in A and 2.5 M in B. The kinetics are given by equation (2) with k_0 = 1.4 x 10^{14} min^{-1} and E = 22 kcal mol^{-1}. The heat of reaction is –211 kJ mol^{-1}, and the heat capacity of the fluid is 1.3 cal cm^{-3} K^{-1}. The reactor is going to be started up at a space time of 10 min by initially filling it with solvent (no reactants or products) at 50 °C and then starting the feed flow. Calculate and plot the outlet molar flow of Y and the outlet temperature as functions of time.

$$2A + B \rightleftarrows 2Y + 2Z \tag{1}$$

$$r_1 = k_1 C_A \tag{2}$$

Assignment Summary

Reaction

$$2A + B \rightleftarrows 2Y + 2Z \tag{1}$$

Rate Expression

$$r_1 = k_1 C_A \tag{2}$$

Reactor: liquid-phase, adiabatic, transient CSTR

Given Constants: V = 1.5 L, T_{in} = 50 °C, $C_{A,in}$ = 1.5 M, $C_{B,in}$ = 2.5 M, $k_{0,1}$ = 1.4 x 10^8 min^{-1}, E_1 = 30 kcal mol^{-1}, ΔH_1 = –211 kJ mol^{-1}, $\breve{C}_p$ = 1.3 cal cm^{-3} K^{-1}, τ = 10 min, and T_0 = 50 °C.

Deliverables: $\dot{n}_Y$ vs. t and T vs. t as graphs

Completion Strategy

This assignment asks for the transient response of a CSTR. Solving the CSTR design equations will yield the outlet molar flow rates and temperature as function of elapsed time.

Formulation of the Calculation

CSTR Design Equations:

$$\frac{d\dot{n}_A}{dt} = \frac{\dot{V}}{V}\left(\dot{n}_{A,in} - 2r_1V - \dot{n}_A\right) \tag{3}$$

$$\frac{d\dot{n}_B}{dt} = \frac{\dot{V}}{V}\left(\dot{n}_{B,in} - r_1V - \dot{n}_B\right) \tag{4}$$

$$\frac{d\dot{n}_Y}{dt} = \frac{\dot{V}}{V}\left(2r_1V - \dot{n}_Y\right) \tag{5}$$

$$\frac{d\dot{n}_Z}{dt} = \frac{\dot{V}}{V}\left(2r_1V - \dot{n}_Z\right) \tag{6}$$

$$\frac{dT}{dt} = -\frac{\dot{V}\breve{C}_p\left(T - T_0\right) + r_1V\Delta H_1}{V\breve{C}_p} \tag{7}$$

CSTR Model Variables: t, $\dot{n}_A$, $\dot{n}_B$, $\dot{n}_Y$, $\dot{n}_Z$, and T

Additional Computable Unknowns: $\dot{V}$, $\dot{V}_{in}$, $\dot{n}_{A,in}$, r_1, k_1, C_A, $\dot{n}_{B,in}$

$$\dot{V} = \dot{V}_{in} \tag{8}$$

$$\dot{V}_{in} = \frac{V}{\tau} \tag{9}$$

$$\dot{n}_{i,in} = \dot{V}_{in}C_{i,in} \qquad i = A \text{ and } B \tag{10}$$

$$k_1 = k_{0,1} \exp\left(\frac{-E_1}{RT}\right) \tag{11}$$

$$C_A = \frac{\dot{n}_A}{\dot{V}} \tag{12}$$

IVODE Solver Requirements:

1. Initial values

$$t\Big|_{t=0} = 0 \tag{12}$$

$$\dot{n}_i\Big|_{t=0} = 0 \tag{13}$$

$$T\Big|_{t=0} = T_0 \tag{14}$$

2. Stopping criterion

$$t = t_f \tag{15}$$

3. Derivatives function that
   a. receives t, $\dot{n}_A$, $\dot{n}_B$, $\dot{n}_Y$, $\dot{n}_Z$, and T at the start of an integration step
   b. calculates the additional unknowns using equations (2), and (8) - (12)
   c. evaluates and returns the design equation derivatives using equations (3) - (7)

Implementation of the Calculations

The Python script, practice_12.py, and Matlab function, practice_12.m, that were written to perform the calculations accompany this solution.

Results and Discussion

The calculations were performed as described above. The outlet molar flow rate of reagent Y is plotted as a function of time in Figure 1, and the outlet temperature vs.

time is plotted in Figure 2. Both quantities smoothly transition to their steady-state values in approximately 50 minutes.

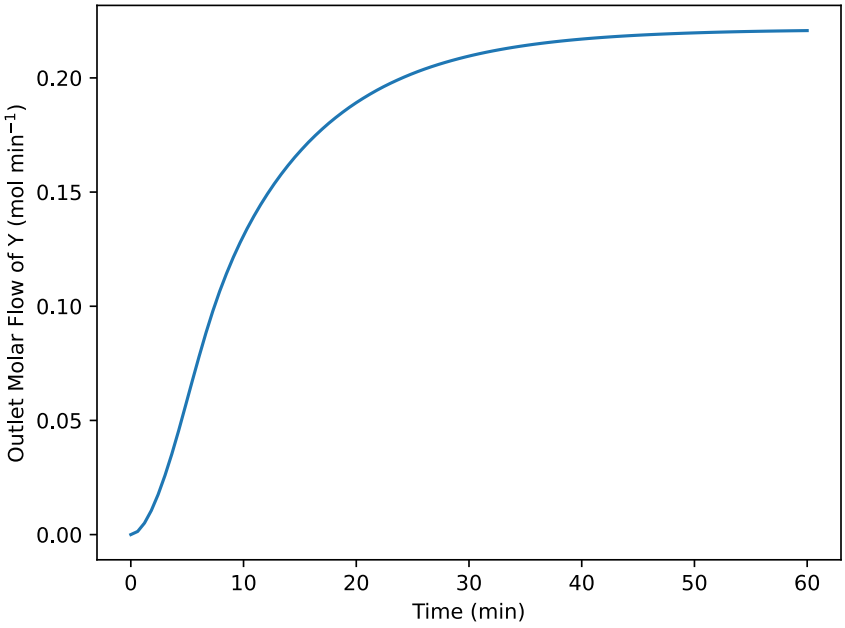


Figure 1. Outlet molar flow rate of reagent Y during CSTR startup.

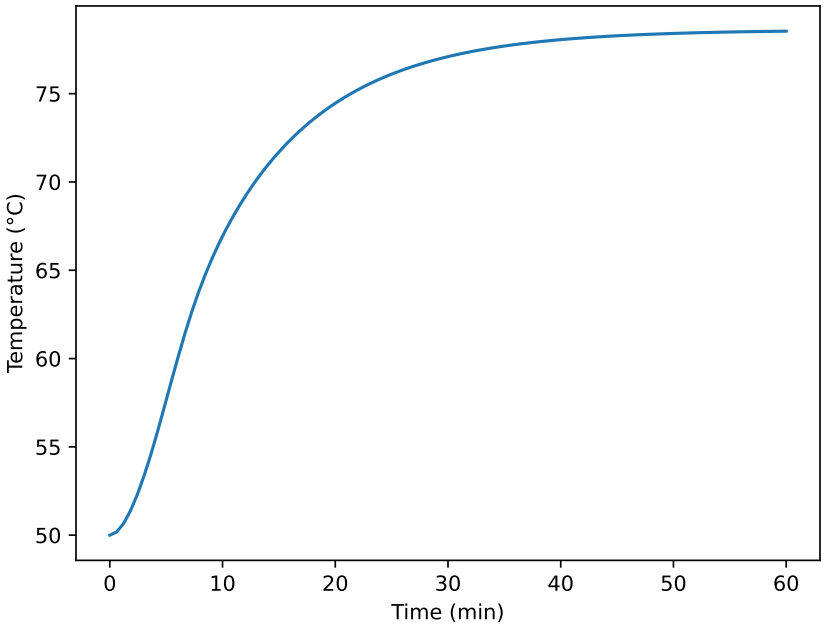


Figure 2. Outlet temperature during CSTR startup.

Before implementing this startup procedure the number of possible steady states should be determined to make sure the desired one is reached. Other startup procedures should also be investigated to see if the time required can be reduces without increasing costs.

Summary

The specified startup process reaches a steady state where the outlet flow of reagent Y is 0.22 mol min^{-1} and the temperature is 77 °C in approximately 50 minutes.